Sequences — problem 9.47

A. Engel — Problem-Solving Strategies · 1998

(IMO 1993.) Let $n > 1$ be an integer. There are n lamps $L_0, \ldots, L_{n-1}$ arranged in a circle. Each lamp is either ON or OFF. A sequence of steps $S_0, \ldots, S_i, \ldots$ is carried out. Step S_j affects the state of L_j only (leaving the states of all other lamps unaltered) as follows :

If L_{j-1} is ON, S_j changes the state of L_j from ON to OFF or from OFF to ON ;

If L_{j-1} is OFF, S_j leaves the state L_j unchanged.

The lamps are labeled mod n, that is, $L_{-1} = L_{n-1}$, $L_0 = L_n$, $L_1 = L_{n+1}$. Initially all lamps are ON. Show that

(a) there is a positive integer $M(n)$ such that after $M(n)$ steps all the lamps are ON again ;

(b) if n has the form 2^k, then all lamps are ON after $(n^2 - 1)$ steps ;

(c) if n has the form $2^k + 1$, then all the lamps are ON after $(n^2 - n + 1)$ steps.